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Human–large language model collaboration in clinical medicine: a systematic review and meta-analysis



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Human–AI collaboration (H + AI) using large language models (LLMs) offers a promising approach to enhance clinical reasoning, documentation, and interpretation tasks. Following PRISMA 2020 (PROSPERO registration: CRD420251068272), we systematically compared H + AI with human-only (H) workflows, searching four databases through June 28, 2025. Ten peer-reviewed studies met eligibility criteria, with three preprints informing sensitivity analyses only. Diagnostic/interpretation accuracy ($k = 2$) showed a positive trend for H + AI (Risk Ratio [RR] 1.59), but was statistically imprecise and non-significant (95% CI 0.08 to 32.74), with 95% prediction intervals (PI) crossing the null. Composite diagnostic/management scores ($k = 2$) showed a statistically significant improvement (Mean Difference [MD] +4.88 percentage points, 95% CI +0.65 to +9.12), yet the PI (−31.65 to 41.42) indicates high real-world uncertainty. Time efficiency ($k = 3$) showed no overall difference (MD +0.4 min, 95% CI −4.18 to +4.97; $I^2 = 70.1\%$). While documentation quality improved, but factual error rates remained high (~26–36%), undermining quality gains. In three-arm settings, H + AI did not universally outperform AI-only. Evidence remains preliminary yet highly uncertain and context-dependent. We recommend preregistered, pragmatic, multicenter trials embedded in real workflows, with harmonized core outcomes that prioritize safety/error metrics and interfaces that surface uncertainty and support verification.

Artificial intelligence (AI)—particularly large language models (LLMs) such as GPT-4, Claude, and task-specific systems like AMIE—is advancing rapidly in healthcare and is widely viewed as a potential paradigm shift in service delivery^{1–7}. Early evidence shows strong performance in standardized examinations, clinical data interpretation, extraction of salient history elements, and initial diagnosis^{1,2,8–20}. In parallel, increasing numbers of institutions are piloting or deploying AI tools for triage, clinical decision support, and documentation, reinforcing a “collaboration-first” narrative: rather than replacing physicians, many organizations are adopting human–AI collaboration (H + AI) models (co-pilot/centaur) that leverage AI’s retrieval, computation, and information-integration capabilities to augment clinical judgment while preserving physicians’ ultimate

authority^{21–28}. Such collaboration is posited to improve efficiency and accuracy and to align more closely with regulatory and ethical expectations for explainability and accountability²⁹.

Despite this momentum, most research has emphasized models’ standalone diagnostic capability and head-to-head comparisons with clinicians. Applications—and reported advantages—span multiple specialties, including emergency and general internal medicine^{2,30}, medical imaging^{31,32}, ophthalmology^{33,34}, gastrointestinal endoscopy^{35,36}, electrocardiography³⁷ and radiology³⁸. A meta-analysis of 83 studies reported an overall diagnostic accuracy of 52.1% for generative AI, with no significant difference versus “overall physicians” or “non-expert physicians,” but significantly lower accuracy than expert physicians³⁹. However, a critical gap remains: direct

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comparisons of human + AI (H + AI) versus human-only (H) and/or AI-only modes on clinical task performance and related outcomes are limited^{23,24,40,41}. Another meta-analysis synthesizing 106 experimental studies (370 effect sizes) found that, on average, human-AI teams performed significantly worse than the best single agent—either human or AI (Hedges' $g = -0.23$; 95% CI -0.39 to -0.07)—underscoring the variability of collaborative effectiveness⁴². Such comparisons are essential for judging the real-world value and generalizability of LLMs in clinical practice. Early evidence from high-quality randomized controlled trials (RCTs) is mixed: some studies found no significant net benefit of AI assistance⁴², others showed clear advantages only in specific tasks and contexts², and several reported a paradox in which AI alone outperformed human + AI teams under certain conditions^{1,2,42–44}.

This tension between theoretical expectations and empirical findings highlights persistent uncertainty about how to collaborate and when collaboration adds value^{25,45,46}. Accordingly, this systematic review seeks to synthesize and rigorously appraise evidence directly comparing human + AI (H + AI) with human-only (H) and/or AI-only modes, moving beyond the a priori assumption that collaboration is inherently beneficial. Our goal is to identify the clinical tasks, interaction and workflow designs, and implementation conditions under which

human-AI collaboration succeeds—and when it may fail—thereby providing actionable, evidence-based guidance for prudent and effective deployment of AI in clinical care.

Results

Literature search and selection

We initially identified 1235 records, including 182 duplicates. After title/abstract screening, 958 records were excluded as irrelevant, and 95 full-text articles were assessed for eligibility. Ten peer-reviewed studies met inclusion criteria for the main analysis, and three medRxiv preprints were added as sensitivity evidence (Fig. 1). We excluded purely observational methodological reports, technical evaluations without clinical outcomes, and studies lacking a control group.

Study characteristics

Of the 10 peer-reviewed studies, eight used a two-arm design (H + AI vs. H), and two additionally included an AI-only arm (not pooled in the primary meta-analysis). Table 1 summarizes key features. Clinical tasks encompassed diagnostic reasoning, triage/Intensive Care Unit differential diagnosis, radiologic and neuro-electrodiagnostic interpretation, clinical documentation, and cross-disciplinary communication. All interventions

Fig. 1 | PRISMA 2020 flow diagram. Study selection for the systematic review and meta-analysis. Records identified through database searching (MEDLINE, Embase, Cochrane Library, Web of Science) through 28 June 2025; numbers shown for screening, full-text assessment, exclusions (with reasons), and studies included in quantitative synthesis. Deduplication, dual-review with third-reviewer adjudication; full strategies in Supplementary Table 4; study-level decisions in Supplementary Table 5.

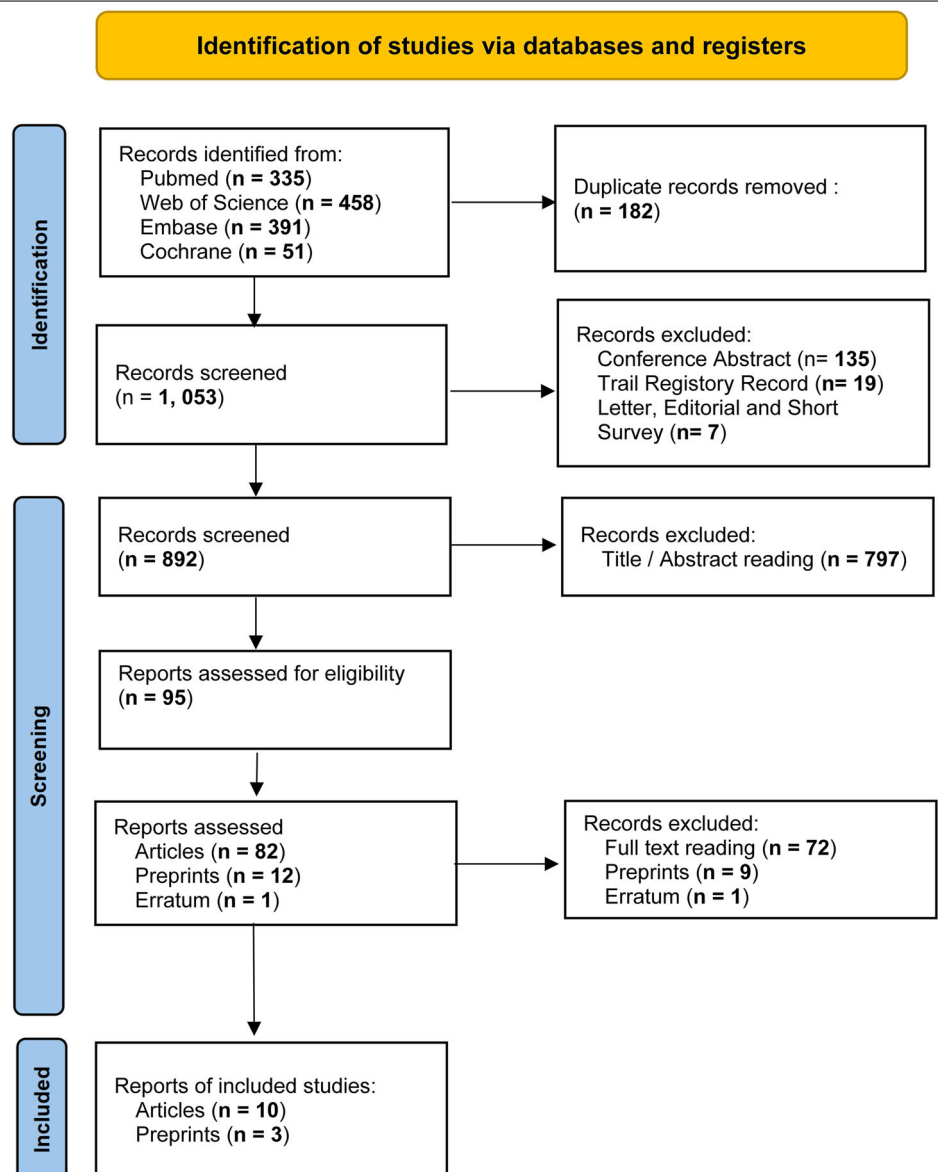


Table 1 Characteristics of included studies						
Study (year; journal)	Design / Arms (main comparison)	N*	Population/setting	Clinical task	LLM / interface	Primary outcome(s)
Baker 2023; JAAOS	3-arm blinded crossover; H + AI(ChatGPT) vs. H(typing/dictation)	11	Ortho clinicians/learners	HPI documentation (standardized pts)	ChatGPT	PDQL-9; time; hallucination check
Goh 2025; Comm Med	Pre-post, single-blind; baseline H → H + AI	50	Licensed physicians	Chest-pain triage (video vignettes)	GPT-4 assist	Guideline accuracy; bias/fairness metrics
Goh 2024; JAMA Netw Open	Parallel RCT, single-blind; Physician +LLM vs. Physician	50	Attendings/residents	Diagnostic reasoning (6 vignettes)	ChatGPT Plus (GPT-4)	Composite reasoning score; time; final diagnosis
Taylor 2025; JAMA Ophthalmol	Prospective randomized QI; LLM-PLS vs. usual notes	362	Non-ophthalmology clinicians (readers)	Cross-disciplinary understanding of notes	LLM-generated PLS	Understanding/clarity; workflow/usability
Goh 2025; Nat Med	Parallel RCT (also AI-only benchmark); H + AI vs. H	92	Attendings/residents	Multistep management (5 cases)	GPT-4 (ChatGPT)	Total performance (0–100); time
Wu 2025; Crit Care	Randomized comparative (AI-only arm reported); AI-assist vs. Non-AI	32	Residents; 48 cases	Critical-illness DDX	DeepSeek-R1	Top-1 accuracy; DDX quality; time
McDuff 2025; Nature	Randomized assistive reader (AMIE-only reported); H + AI vs. H	20	Generalist physicians	DDx on NEJM CPC cases	AMIE (PaLM-2-L) chat	Top-10 DDX accuracy; appropriateness/completeness; time
Gorenshtein 2025; J Neurol	Prospective RCT (workflow); AI-assist vs. Physician-only	3 (200 cases)	Neuromuscular clinicians	EDX/EMG reporting	LLM-assisted tool	AI GERS/report quality; usability/trust; time
Yu He Ke 2025; NPJ Digit Med	Randomized crossover (real clinic); H + AI(PEACH) vs. H	14 residents; 272 visits	Pre-op clinic	Pre-op assessment & note	PEACH LLM-CDSS	Primary: documentation time/visit; Secondary: quality, errors/safety, acceptance, economics
Kim 2025; Eur Radiol	Retrospective reader/usability; Reader +LLM vs. Reader	6	Radiology residents	Brain MRI DDX	LLM-assisted reader (n/s)	DDx accuracy/efficiency; usability

*N denotes the number of clinician participants unless otherwise specified. AI GERS AI-Generated Evaluation of Report Standard, AI artificial intelligence, CDSS clinical decision support system, CPC clinicopathological conference, DDX differential diagnosis, EDX/EMG electrodiagnostic/electromyography, H human-only, H + AI human-AI collaboration, HPI/history of present illness, LLM/large language model, MRI/magnetic resonance imaging, n/s not specified, NEJM New England Journal of Medicine, PaLM/Pathways Language Model, PDQL-9 Physician Documentation Quality Instrument (9-item), PLS plain-language summary, RCT randomized controlled trial.

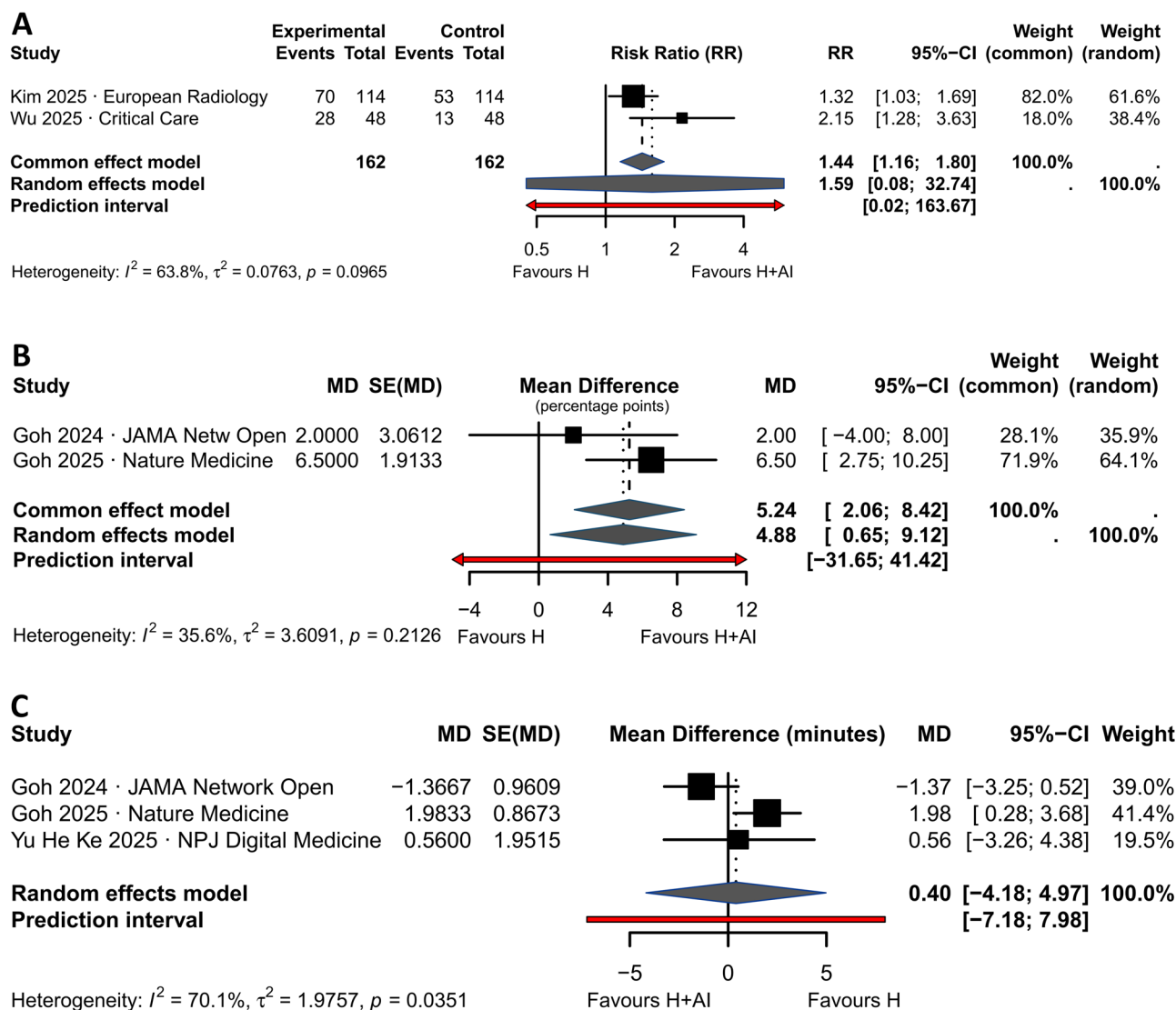


Fig. 2 | Meta-analytic effects of human-AI collaboration (H + AI) versus human-only (H). A Diagnostic and interpretive correctness (binary outcomes). Pooled risk ratios (RRs) were calculated using common-effect and random-effects inverse-variance models. **B** Composite diagnostic and management performance (continuous outcomes, percentage points). Pooled mean differences (MDs) are shown from common-effect and random-effects models. **C** Time efficiency (continuous outcomes, minutes). MDs are defined as (H + AI - H); positive values indicate that

the AI-assisted workflow is slower than the human-only workflow. Note: Square sizes are proportional to study weights, and diamonds represent pooled estimates. The vertical reference line indicates no effect (RR = 1 or MD = 0). Red bars represent 95% prediction intervals (PI) derived from random-effects models. Between-study variance (τ^2) was estimated using the Paule-Mandel estimator, and all random-effects models utilize Hartung-Knapp-Sidik-Jonkman (HKSJ) adjusted confidence intervals to account for the small number of studies ($k = 2$ or 3).

employed LLM-based systems—predominantly GPT-4/ChatGPT, with AMIE, DeepSeek-R1, and PEACH also represented. Participants were mainly attending physicians and residents; a minority of studies enrolled medical students or subspecialty readers. Primary outcomes included diagnostic/interpretive accuracy, composite reasoning scores, time efficiency, documentation/report quality (PDQI-9, AIGERS), and cross-disciplinary understanding.

Meta-analytic results

Accuracy/quality performance. Two peer-reviewed studies ($k = 2$; Kim⁴⁷, Wu⁴³) reported event-type data suitable for pooling. Study-level Risk Ratios (RRs) were 1.32 (95% CI 1.03–1.69) and 2.15 (95% CI 1.28–3.63). The common-effect (fixed-effect) model yielded an RR of 1.44 (95% CI 1.16–1.80). Our primary random-effects model (Paule-Mandel with HKSJ adjustment, appropriate for $k = 2$) yielded an RR of 1.59 (95% CI 0.08–32.74; $I^2 = 63.8\%$; $\tau^2 = 0.0763$; 95% PI 0.02–163.67) (Fig. 2A).

A paired reader-case study not included in the meta-analysis—McDuff¹ (Nature; NEJM CPC case set)—showed concordant, statistically significant gains (Top-1: 29.2% → 59.1%; Top-10: 33.6% → 51.7%; McNemar $P < 0.01$) and is cited as external consistency. Because its paired-variance structure differs from that of parallel RCTs, it was not combined with independent-sample trials. In a preprint sensitivity analysis (Spitzer)⁴⁸, accuracy improved from 77.6% (H) to 89.6–95.8% (H + AI) across prompting strategies within the subset receiving consistent AI advice, aligning in direction; nevertheless, unavailable variance parameters precluded updating the random-effects point estimate or interval. Overall, while the point estimate favors H + AI, the 95% CI includes the null (0.99–2.70) and the 95% PI crosses it (0.78–3.26). Therefore, the direction of effect is highly uncertain, and high heterogeneity and imprecision severely constrain extrapolation across settings.

Composite diagnostic/management percentage scores. Two parallel RCTs ($k = 2$; Goh², Goh⁴⁹) reported composite percentage-score

outcomes. Individual effects were +2.0 percentage points (95% CI, −4.0 to +8.0) and +6.5 percentage points (95% CI, 2.75 to 10.25); domain-specific effects were +12.1% for diagnosis and +6.1% for management. Random-effects MD was +4.88 percentage points (95% CI +0.65 to +9.12; $I^2 = 35.6\%$; $\tau^2 = 3.6091$; 95% PI −31.65 to 41.42); the fixed-effect MD was 5.24 (95% CI 2.06–8.42). Effects suggest small-to-moderate improvements with scale heterogeneity and small k limiting precision (Fig. 2B).

A preprint sensitivity analysis (Everett 2025) showed improvements of +6.8% and +9.8% versus conventional resources (both $P < 0.0004$), with magnitudes within the range of the main analysis; owing to heterogeneity in score construction and insufficient dispersion data, these effects were not formally pooled⁵⁰. Overall, findings indicate a small-to-moderate improvement, with prediction-interval instability attributable to scale heterogeneity and the small number of studies.

Time/efficiency. Three studies (Goh⁴⁹, Goh², Ke⁵¹) were pooled as mean difference (minutes; H + AI − H). The random-effects MD was +0.40 min [95% CI −4.18 to +4.97; $I^2 = 70.1\%$, $\tau^2 = 1.9757$, Cochran's Q ($df = 2$) = 6.70, $p = 0.0351$; 95% PI −7.18 to 7.98] indicating no overall difference and context-dependent effects (Fig. 2C). By study, Goh⁴⁹ reported a modest delay (+119 s; 95% CI, +17 to +221), Goh² reported a non-significant speed-up (−82 s; 95% CI, −195 to +31), and the real-world PEACH (Perioperative AI CHatbot) study⁵¹ showed a near-null overall effect (+0.56 min; 95% CI, −3.27 to +4.38) but significant speed-ups in moderately complex cases and among more senior physicians. Wu 2025⁴³ reported a median reduction from 32.0 to 16.2 minutes but was not pooled owing to missing variance estimates.

A preprint sensitivity analysis (Castaño-Villegas 2025) suggested roughly halved response times (~3 min faster) and ~66% fewer searches⁵²; however, large task/workflow differences limited comparability, and its inclusion would not materially change the near-zero overall estimate or reduce heterogeneity. The observed heterogeneity appears strongly driven by the study setting. Notably, “near-clinical” simulations showed mixed results^{24,49}, while the real-world pragmatic, workflow-embedded study⁵¹ showed a near-null overall effect. This suggests efficiency gains observed in simulated tasks may not translate directly to complex clinical workflows.

Additional qualitative evidence and consistency. Gorenshstein⁵³ (EMG/EDX) found H + AI $\approx$ H [mean (SD) AIGERS 0.94 (0.11) vs. 0.94 (0.13); $P = 0.32$], whereas AI-only performed markedly worse than H [mean (SD) 0.70 (0.26); $P < 0.001$], suggesting that collaboration is not inferior to human-only performance, while AI alone may reduce quality. In Baker⁵⁴ (HPI documentation), H + AI improved PDQI-9 but incurred ~36% factual errors, highlighting a potential dissociation between documentation structure/quality and factual correctness. Tailor⁵⁵ (plain-language summaries) reported improvements in understanding, satisfaction, and clarity (understanding +9.0 percentage points; 95% CI 0.3–18.2) with ~26% errors (mostly low risk). Goh⁴⁹ suggested improved guideline concordance and selected fairness metrics but provided stratified differences without denominators, precluding meta-analysis. These observations are directionally consistent with the primary results, but evidence grading and generalization should distinguish “near-clinical” from “real-clinical” settings and account for task and scale comparability.

AI-only comparator arms. In Wu⁴³, AI-only Top-1 accuracy was ~60% (statistically similar to H + AI at 58%), whereas in Gorenshstein⁵³, AI-only report quality was significantly worse than both H and H + AI. These findings provide evidence that collaboration is not inherently superior to AI-only across all tasks; observed gains likely depend on interface design, user training, and workflow embedding.

Sensitivity and special-design analyses. The robustness of the primary findings was confirmed across several sensitivity analyses. Pooled estimates remained insensitive to variations in the τ^2 estimator (e.g.,

Paule–Mandel, REML) or the specific effect metric (e.g., RR). This stability extended to the primary pooled outcomes: for both diagnostic accuracy ($k = 2$) and composite scores ($k = 2$), the 95% prediction intervals consistently crossed the null, reinforcing high uncertainty and cautioning against overgeneralization. Similarly, all specifications for time/efficiency ($k = 3$) analyses confirmed no overall difference and substantial, task-driven heterogeneity.

Two key special-design analyses provided further context. First, analysis of the included three-arm studies (H + AI vs. AI-only) yielded a synergy ratio of approximately 1, providing no statistical evidence that H + AI collaboration inherently outperforms the AI-only agent. Second, a qualitative assessment of three excluded preprints found their results directionally concordant with our main analysis: positive trends for accuracy (Spitzer)⁴⁸ and composite scores (Everett)⁵⁰, and high, context-dependent variability for time (Castaño-Villegas)⁵². Crucially, the qualitative inclusion of these preprints did not materially alter the main pooled estimates, heterogeneity, or the low-certainty GRADE ratings, confirming the robustness of our uncertain conclusions.

Finally, due to small- k constraints ($k < 10$ for all outcomes), formal tests for publication bias (e.g., funnel plots, Egger's test) were not performed as they would lack statistical power. Instead, we relied on concordance between fixed- and random-effects estimates and the inclusion of null/negative effects in the time-efficiency analysis; this qualitative assessment suggested a low-to-moderate risk of positive publication bias.

Risk of bias, reporting bias, and GRADE. Risk of bias (RoB 2/ROBINS-I). Of the 10 studies, 9 RCTs were appraised with RoB 2 and one non-randomized study with ROBINS-I. Overall, 2/9 RCTs were at low risk and 7/9 had some concerns; the nonrandomized study was at moderate risk (Fig. 3; Supplementary Table 1 and 2). Key RCT issues involved D2 (deviations from intended interventions/adherence shortfalls), D4 (subjective outcome measurement with incomplete blinding), and D1/D5 (insufficient modeling of reader–case clustering and crossover sequence effects). For the nonrandomized study, major risks arose from confounding and participant selection; lack of preregistration raised concerns about selective reporting.

Reporting bias and small-study effects. Given $k < 10$ for each pooled outcome, we did not perform funnel plots or Egger/Begg tests. Instead, we assessed concordance between fixed- and random-effects estimates ($RR \approx 1.45$ – 1.63 ; $MD \approx +4.9$ percentage points), inclusion of near-zero/negative time effects, and agreement in direction and magnitude between preprints and peer-reviewed studies. Together, these suggest a low-to-moderate risk of publication bias. Residual concerns relate to outcome selection, heterogeneity in scale construction, and incomplete reporting of clustering/sequence effects.

GRADE (Supplementary Table 3). Performance accuracy/quality (binary, $k = 2$): Low certainty, downgraded for risk of bias, inconsistency, and imprecision (CI crossing the null). Composite diagnostic/management percentage scores (continuous, $k = 2$): Moderate certainty, downgraded for indirectness (non-identical composite construction) and imprecision (small k). Time/efficiency (continuous, $k = 3$): Very low/low certainty, downgraded for D2/adherence/non-ITT issues, high heterogeneity, indirectness (task/workflow variability), and imprecision. Sensitivity analyses incorporating preprints did not change these ratings.

Discussion

As a systematic synthesis of evidence on large-language-model (LLM)-enabled human–AI collaboration (H + AI), we included 10 trials and observed the following. For diagnostic and management performance, two combinable studies yielded a random-effects RR of 1.59 (95% CI 0.08–32.74; $I^2 = 63.8\%$), using the HKSJ adjustment appropriate for small k . This estimate, while directionally positive, is statistically non-significant and highly uncertain, with a 95% CI that broadly crosses the null. Furthermore, the 95% PI (0.02–163.67) is extremely wide, confirming a lack of generalizability. Two parallel RCTs reporting continuous composite scores

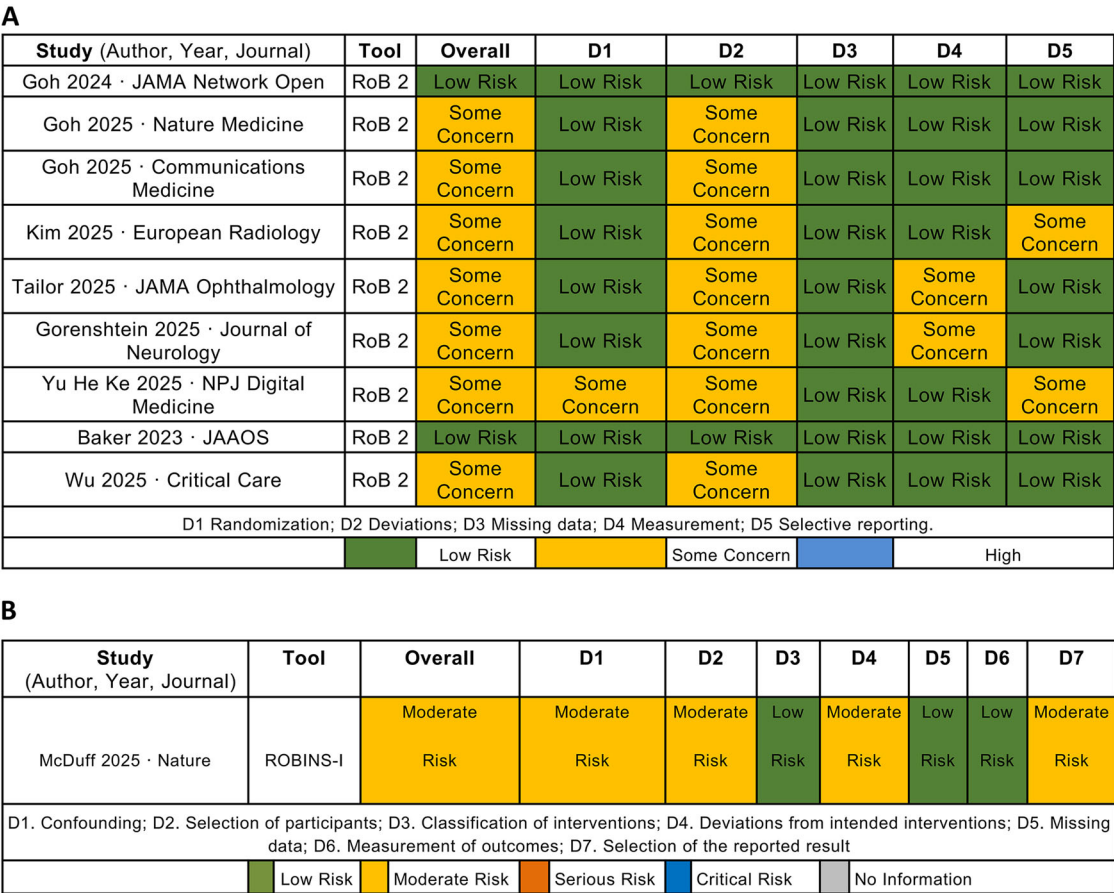


Fig. 3 | Risk of bias (RoB) summary for randomized studies (Panel A) and the non-randomized study (Panel B). **A (RoB 2)** assessed 5 domains: D1 (Randomization process), D2 (Deviations from intended interventions), D3 (Missing outcome data), D4 (Measurement of outcomes), and D5 (Selection of the reported result). Color coding (color-blind friendly): Green = Low Risk; Yellow = Some Concerns; Blue = High Risk (not observed). Overall, 2 trials were at Low Risk, and 7 trials had Some Concerns, driven chiefly by D2 (non-blinding/adherence deviations). **B (ROBINS-I)** assessed 7 domains: D1 (Confounding), D2 (Selection of participants), D3 (Classification of interventions), D4 (Deviations from intended interventions), D5 (Missing data), D6 (Measurement of outcomes), and D7 (Selection of the reported result). Color coding (color-blind friendly): Green = Low Risk; Yellow = Moderate Risk; Orange = Serious Risk; Dark Blue = Critical Risk; Gray = No Information. The overall rating was moderate risk, reflecting concerns in domains D1, D2, D4, and D7. Note: Full domain-level tables appear in Supplementary Table 1 and 2.

produced a pooled MD of +4.88 percentage points (95% CI +0.65 to +9.12; $I^2 = 35.6\%$). While this finding is statistically significant (CI does not cross zero), its clinical significance and robustness are questionable. Clinically, an improvement of 4.88 percentage points could correspond to approximately 5 fewer diagnostic or management errors per 100 clinical encounters. However, this interpretation must be tempered by the small number of studies ($k = 2$) and the extremely wide 95% prediction interval (−31.65 to 41.42), which indicates that future studies could plausibly find no benefit or even substantial harm. For time outcomes, the pooled MD was +0.40 minutes (95% CI −4.18 to +4.97; $I^2 = 70.1\%$), showing no overall difference and substantial sensitivity to task characteristics, depth of interaction, and workflow integration. Considering RoB 2/ROBINS-I (overall “low risk/some concerns,” with “serious” risk in one non-randomized study) and GRADE (moderate certainty for diagnostic and documentation outcomes; low certainty for time), the current best evidence supports a conditional net benefit under specific tasks and implementation conditions, but it is insufficient to justify universal claims. Future work should prioritize preregistered, multicenter parallel RCTs with harmonized primary outcomes and instruments, explicit modeling and correction of clustering and sequence effects, intention-to-treat analyses, and minimized contamination to enhance internal and external validity.

Performance gains from “AI + clinician” are not automatic; rather, they depend on multiple factors, including task attributes, interface and

prompt design, human-factors training, and workflow integration^{42,43,56,57}. Four themes—the collaboration paradox^{1,28}, task specificity^{2,57}, quality-accuracy trade-offs⁵⁸ and the primacy of human factors^{1,49}—collectively indicate that merely placing an AI tool in clinicians’ hands does not guarantee consistent net benefit. Human–AI collaboration is, fundamentally, a complex sociotechnical process.

A critical finding, supported by the three-arm studies^{43,53}, is the “collaboration paradox”: H + AI collaboration does not demonstrate universal superiority over a strong AI-only agent. In Wu⁴³, for example, H + AI accuracy (58%) did not significantly exceed, and was directionally similar to, standalone AI accuracy (~60%). As noted in our sensitivity analysis, the estimated synergy ratio for this task was approximately 1, providing no statistical evidence of true collaborative gain over the AI-only agent. Plausible mechanisms include cognitive dissonance when AI advice conflicts with clinicians’ initial judgments and miscalibration driven by automation and confirmation biases^{59,60}. This paradox has direct implications for clinical workflow and sociotechnical design. It indicates that “human-in-the-loop” is not a universal safeguard, but a complex interaction that can introduce new failure modes—such as anchoring on incorrect AI suggestions or diluting high-performing AI outputs through inconsistent human overrules. Consequently, workflow implementation should move beyond a one-size-fits-all collaboration model toward a task-differentiated strategy. For highly structured, low-variance tasks where AI performance is robustly

superior, strategic delegation to AI-only pathways with targeted “human-on-exception” oversight may be optimal. In contrast, for high-ambiguity, high-stakes scenarios (e.g., complex differential diagnosis), workflows must be deliberately engineered to support effective human–AI teaming, with interfaces that surface model uncertainty⁶¹, provide contrastive and traceable evidence chains⁶², and embed mandatory verification protocols or secondary sign-offs⁶³. Such designs aim to mitigate cognitive bias, reduce vigilance burden, and realize collaborative benefit without compromising safety.

To guide deployment, we propose a 2 × 2 classification framework defined by task complexity (high/low) and task structure (high/low): collaborative value is greatest in the high-complexity, low-structure quadrant (e.g., ambiguous presentations with broad differentials) and minimal in the low-complexity, high-structure quadrant (e.g., templated documentation). This framework can help health systems allocate limited resources to scenarios with the highest clinical value.

Regarding role delineation, model error rates and hallucination risk imply that clinicians should shift from “information generators” to expert verifiers, with core competencies in critically appraising and recalibrating AI outputs. Ethically and legally, ultimate responsibility remains with clinicians despite AI assistance; this supervisory and accountability burden (“vigilance tax”) must be supported by governance, audit, and traceability mechanisms.

Consistent with prior evidence, most current studies adopt head-to-head AI–clinician designs. In standardized interviews, remote OSCEs, or text-based tasks, LLMs can perform at a specialist level and outperform generalists or junior clinicians on several dimensions^{1,9,20,39}; in patient-facing Q&A, independent raters often prefer AI-generated responses for quality and empathy⁶⁴. Cross-disciplinary randomized evidence also shows that plain-language summaries for out-of-specialty readers can markedly improve understanding and communication, suggesting larger relative gains for users with less specialty exposure in collaborative settings^{19,65}. These observations support a strategic shift from “replacement” to collaboration and emphasize that reproducible benefit depends on task selection and workflow redesign, rather than treating LLMs as a passive “second opinion”.

The high heterogeneity we observed arises mainly from three sources: (i) misalignment of task domains and endpoint definitions (e.g., differing definitions of “correctness” for diagnosis/triage vs. documentation, rater composition, and scoring rules); (ii) implementation differences and human-factors considerations (presentation of uncertainty, availability of explanatory evidence, and review thresholds affect calibration and anchoring); and (iii) model generation and prompting (versioning, fine-tuning corpora, and safety guardrails influence hallucination rates and interpretability, with downstream effects on clinical endpoints). Prediction intervals that cross the null further underscore the need for contextual re-evaluation—particularly for time outcomes—which should be revalidated across new services, conditions, and workflows.

Our review has several strengths: explicit delineation of task domains and comparator arms; incorporation of three-arm evidence to disentangle the marginal source of synergy; and application of the ratio-of-means for time outcomes to facilitate cross-study synthesis. A primary limitation of this review is the critical reliance of the included evidence on “near-clinical” simulations (e.g., vignettes, standardized case files) rather than patient-facing trials. This reliance severely limits the external validity and generalizability of our findings, as simulated settings cannot capture the time pressures, interruptions, and workflow complexities of real-world clinical practice. Further limitations include the small number of studies and participants; some outcomes being pooled from only two trials or described narratively; susceptibility to learning and sequence effects in crossover and pre–post designs; inconsistent blinding and control of concomitant interventions; small three-arm samples; and high heterogeneity driven by task and setting variability (e.g., simulated vs. real-world workflows). Despite HKSJ adjustments and multiple τ^2 estimators in small-sample settings, variance estimates may remain unstable; additionally, $k < 10$ limits the power of publication-bias diagnostics.

Practice implications. A critical finding of this review is the high rate of factual errors (i.e., hallucinations), with studies reporting ~26–36% errors (see Results). This highlights a potential dissociation between documentation structure and factual accuracy. Therefore, we strongly recommend that safety-critical metrics (e.g., rate of error detection and correction by the user), not just guideline concordance or efficiency, be prioritized as primary endpoints in phased pilots to avoid trading efficiency for safety. For diagnostic and triage scenarios, guardrails should include uncertainty disclosure, de-anchoring mechanisms, and mandatory review. Roles should be explicit—AI handles retrieval and recall, whereas clinicians ensure calibration and gatekeeping—with tiered enablement and oversight by seniority to stabilize net benefit.

Future research. Pragmatic multicenter trials should embed interventions within real clinical workflows to evaluate real-world efficiency and patient-level outcomes, moving beyond the current “near-clinical” simulations. To address the high heterogeneity observed in this review, these trials must prioritize harmonized outcome definitions and Core Outcome Sets (COS), with a strong emphasis on safety/error endpoints (e.g., hallucination rates)⁶⁶. Dedicated Human-Computer Interaction studies are needed to design interfaces that support traceability and mitigate cognitive biases⁶⁷. Furthermore, longitudinal designs are essential to assess long-term impacts not captured in short-term trials, including clinician skill retention (or de-skilling), cognitive calibration, and cost-effectiveness⁶⁸. Methodologically, preregistration with standardized reporting of these workflow and safety details is imperative^{40,41}. Finally, prospective equity analyses and real-time safety monitoring must be established to ensure technological advances do not exacerbate disparities⁶⁹.

In summary, LLM-enabled human–AI collaboration yields statistically suggestive but highly uncertain, context-sensitive benefits in diagnostic/management and documentation tasks. The pooled estimates for diagnostic and management quality are not robust, with 95% prediction intervals crossing the null, indicating a high risk of no benefit in different clinical settings. In light of GRADE (moderate certainty for diagnostic/management and documentation quality; low for time efficiency), the most prudent path is contextualized, phased deployment under strengthened safety and governance guardrails, coupled with standardized trials, transparent reporting, and continuous monitoring to improve cross-context generalizability and system-level net benefit.

Methods

Protocol and registration

This review adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guideline⁷⁰. The protocol was prospectively registered with PROSPERO (CRD420251068272). Procedures for study identification, deduplication, and screening are summarized in the PRISMA flow diagram (Fig. 1; Supplementary Table 4 and 5).

Search strategy and study selection

We systematically searched MEDLINE (PubMed), Embase, the Cochrane Library, and Web of Science Core Collection from database inception to June 28, 2025. Strategies combined controlled vocabulary and free-text terms (MeSH/Emtree plus text words) related to artificial intelligence and large language models (“artificial intelligence,” “natural language processing,” “large language model/LLM,” “ChatGPT,” “GPT-3.5/4/4o,” “Claude,” “Gemini”) and to clinical applications (clinical diagnosis, decision-making, triage, management, documentation). Field limits and proximity/truncation operators were tailored to each database. Studies were restricted to English-language publications owing to resource constraints.

Eligibility criteria were specified a priori using a PICO framework: (i) Population: licensed clinicians engaged in clinical care; (ii) Intervention: LLM-based AI assistance; (iii) Comparator: standard human-only clinical workflows, which could include non-AI decision support (e.g., guidelines, scoring systems) but no AI components; (iv) Outcomes: the primary outcome was clinical correctness; secondary outcomes included time efficiency, documentation/report quality, and cross-disciplinary understanding/

communication. Studies were required to directly compare a “clinician+AI” arm with either an “AI-only” or “clinician-only” arm. Full search strings appear in Supplementary Table 4.

Data extraction

Peer-reviewed studies formed the basis of the primary meta-analysis. Preprints were used exclusively for sensitivity analyses. Two reviewers independently extracted data using a pre-piloted, calibrated form (Supplementary Table 6). Disagreements were resolved by discussion or by a third reviewer. Extracted items included study design; setting/task; unit of randomization and analysis; comparator structure (primary focus on H + AI vs. H; three-arm trials recorded as H + AI/H/AI); participant profession and sample size; task type and case source; LLM model/interface and interaction mode; and blinding.

To enhance comparability, outcomes were standardized to pre-specified metrics: diagnostic/management performance as percentage-point mean difference (MD_{pctpt}; positive favors H + AI); time efficiency as the log ratio of means (log RoM; back-transformed RoM < 1 favors H + AI); diagnostic accuracy as a log relative effect (preferentially lnRR; > 0 favors H + AI); and documentation quality as Hedges g (> 0 favors H + AI). For three-arm studies, H + AI vs. AI effects were also extracted. Risk of bias was assessed using RoB 2 or ROBINS-I. Complex designs were analyzed at the randomization/analysis unit; studies with critical missing data were summarized narratively.

Risk of bias assessment

Randomized controlled trials were appraised with the Cochrane RoB 2 tool⁷¹, non-randomized studies were assessed with ROBINS-I⁷². Full domain-level judgments appear in Supplementary Table 1 and 2.

Data synthesis and statistical analysis

For each outcome domain, we reported both common-effect (fixed-effect) and random-effects results in parallel, mirroring the forest plots (“Common effect model,” “Random effects model,” “Prediction interval”). Random-effects models used inverse-variance weighting with the between-study variance (τ^2) estimated by the Paule–Mandel method. Given the small number of studies ($k \leq 3$) for all pooled outcomes, we adopted the Hartung–Knapp–Sidik–Jonkman (HKSJ) adjustment as the primary method for calculating confidence intervals to ensure a robust and conservative variance estimation. To reflect cross-setting generalizability, we reported 95% prediction intervals (PI) for all random-effects syntheses¹⁹. Statistical heterogeneity was summarized by I^2 , τ^2 , and the Cochran’s Q test [reported as Q(df, p-value)]; PI width was used as a complementary descriptor of external heterogeneity.

Effect measures. Our analytical strategy prioritized mean differences (MD) for continuous outcomes measured on the same scale (e.g., percentage points, minutes) to enhance clinical interpretability. For studies reporting medians and interquartile ranges (IQR) or ranges, we estimated means and standard deviations (SDs) using the methods described by Luo et al.⁷³ and Wan et al.⁷⁴; studies for which this data harmonization was not feasible were synthesized narratively. Based on this strategy, for diagnostic/interpretive accuracy (binary; Fig. 2A), the effect was the risk ratio (RR), with study-level log RRs and standard errors computed from 2×2 tables (common-effect: Mantel–Haenszel; random-effects: inverse-variance with Paule–Mandel τ^2). For composite diagnostic/management percentage scores (continuous; Fig. 2B), the primary metric was the mean difference in percentage points (MD, p.p.). For time (continuous; Fig. 2C), the primary metric was the MD in minutes (defined as H + AI–H), pooled using random-effects (Paule–Mandel) with HKSJ-adjusted CIs and a 95% PI. In sensitivity analyses, we explored alternative metrics, including standardized mean differences (Hedges’ g) to assess scale heterogeneity and the log ratio of means (log RoM) for time outcomes.

Given the small k per outcome, subgroup assessments were descriptive only; no inferential meta-regression and no funnel/Egger/Begg tests were performed. Robustness was examined with estimator switching

(conventional vs. HKSJ CIs). Certainty of evidence was graded using GRADE⁷⁵. All statistical tests were two-sided, with an α level set at 0.05. Analyses were performed in R version 4.5.2 (R Foundation for Statistical Computing, Vienna, Austria) using the meta (version 8.2-1) package.

Data availability

The raw data extracted from the included studies to support the findings of this systematic review are provided in Supplementary Table 6.

Code availability

The R code used to perform the meta-analyses and generate Fig. 2 is publicly available on Zenodo at: <https://doi.org/10.5281/zenodo.17611904>.

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Author contributions

X.Y., Y.L., Y.H., and Z.Q. conceived and designed the study. G.W., K.Z., J.J., C.W., H.B., and H.L. collected and verified the data and curated the clinical materials. G.W. and K.Z. performed data processing and statistical analyses and drafted the manuscript. X.Y., Y.L., Y.H., and Z.Q. critically revised the manuscript. All authors interpreted the results and approved the final version.

Competing interests

The authors declare no competing interests.

Additional information

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